

- 2 (a) Fig. 2.1 shows a simplified model of a man lifting a suitcase. The spine can be treated as a rigid rod.



Fig. 2.1

When the suitcase is lifted, three forces P , Q , and R act on the spine which is in equilibrium. P is the force due to lifting the suitcase, Q is the force exerted by the muscles and R is the force acting on the base of the spine.

Fig. 2.2 shows the directions and points of action of only P and Q .

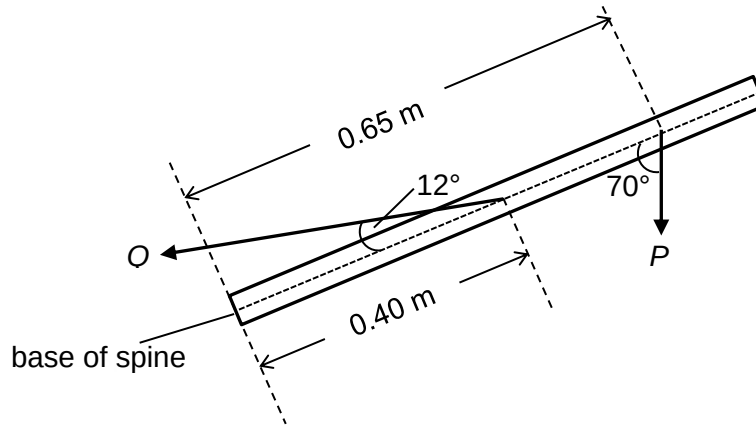


Fig. 2.2

- (i) State the conditions required for a body to be in equilibrium.

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- (ii) Force R acts at the base of the spine.

On Fig. 2.3, draw an arrow to represent the direction of force R .

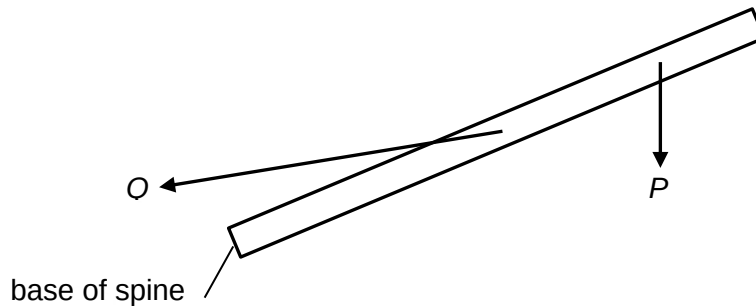


Fig. 2.3

[1]

- (iii) Calculate the ratio of $\frac{Q}{P}$.

$$\frac{Q}{P} = \dots\dots\dots [2]$$

- (b) The suitcase of mass 11 kg is tied to a light rope that is passed over a pulley as shown in Fig. 2.4. When the man applies a force of 30 N on the rope, the suitcase accelerates horizontally across a rough floor. The frictional force between the suitcase and the rough floor is 8.5 N.

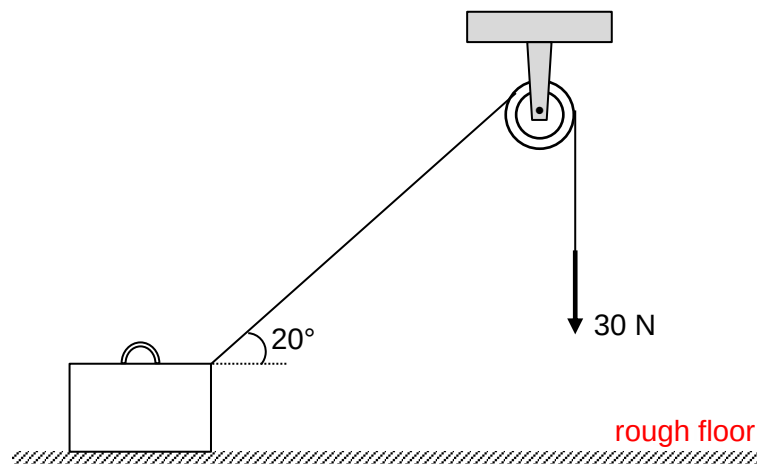


Fig. 2.4

- (i) Determine the normal contact force acting on the suitcase.

normal contact force = N [2]

- (ii) State Newton's second law of motion.

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..... [1]

- (iii) Determine the horizontal acceleration of the suitcase.

horizontal acceleration = m s^{-2} [2]